
Adversarial Attacks and Defenses for Traffic Sign Classification

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Abstract

Traffic sign recognition is a critical safety component of autonomous driving, yet image classifiers are vulnerable to nearly imperceptible adversarial perturbations that can lead to dangerous misclassifications. In this project, We will study both white-box and gray-box attacks on a traffic sign classifier using existing attack methods. In addition, we propose a custom edge-aware attack built on top of these methods and evaluate multiple defenses, including preprocessing, adversarial training, morphology-based denoising, and FFT-based filtering.

1 Introduction

Traffic sign classification is a critical vision task in autonomous driving and driver assistance systems. Misclassifications under adversarial perturbations, such as failing to recognize a stop, speed limit or warning signs can lead to unsafe driving decisions. Due to its safety-critical nature, building robustness against adversarial attack is an important machine learning problem.

We will first train a baseline classifier on clean traffic-sign images and then evaluate the classifier under several adversarial threat models. We will consider both the white-box setting, where the attacker has full access to the model, and the gray-box setting, where the attacker has only partial or imperfect model knowledge. Next, we will evaluate a range of defense strategies and analyze the trade-off between clean performance and robustness to adversarial attacks.

2 Dataset and Features

We will base our experiments on the the German Traffic Sign Recognition Benchmark (GTSRB) (1), a real-world image dataset created for the task of traffic sign classification. During training, we will resize images to a fixed resolution, normalize pixel values, and optionally apply light data augmentation.

Our baseline classifier will be ResNet-18 (2). The main features used by the model are visual patterns in the image, including color, shape, texture, and especially boundary information. Since traffic sign classification strongly depends on contour and edge cues, we are particularly interested in attacks and defenses that use edge-related information.

3 Methods

We will first train a baseline classifier on clean images using cross-entropy loss and standard gradient-based optimization. After obtaining the clean baseline, we will generate adversarial examples under both white-box and gray-box threat models. In the white-box setting, the attacker has full access to the model parameters and gradients. In the gray-box setting, the attacker only has approximate knowledge of the target model. We will compare how strongly attacks transfer from the approximate model to the target model.

3.1 Adversarial Attack

For attack generation, we will use FGSM (3), PGD (4), and the Carlini & Wagner attack (5). FGSM perturbs the image once in the gradient sign direction,

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x L(f_\theta(x), y)),$$

while PGD applies this update iteratively and projects the perturbed image back to the allowed ϵ -ball. We will also implement a custom *edge-aware attack*. First, we compute an edge map $E(x)$ from a lightly Gaussian-smoothed image and normalize it to $[0, 1]$. Then we define a weight map

$$W(x) = 1 + \lambda E(x),$$

so that pixels on strong edges receive larger perturbations. Using this weight map, the attack update becomes

$$x_{\text{adv}}^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)} \left(x_{\text{adv}}^t + \alpha W(x) \odot \text{sign}(\nabla_x L(f_\theta(x_{\text{adv}}^t), y)) \right).$$

This lets us test whether traffic sign classifiers are especially sensitive to perturbations concentrated on edges and boundaries. We will generate adversarial datasets using FGSM, PGD, and custom edge-aware attack, and then evaluate the baseline model on each dataset.

3.2 Defense Training

For defense methods, we will compare: (1) no defense, (2) preprocessing-based defenses (such as Gaussian blur, median filtering, and JPEG compression), (3) adversarial training, and (4) custom denoising methods. In adversarial training, we generate adversarial examples during training and update the model using adversarial loss, which can be viewed as the robust optimization problem

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\| \leq \epsilon} L(f_{\theta}(x + \delta), y) \right].$$

As custom defenses, we will test morphology-based denoising and FFT-based denoising. For morphology-based denoising, we will apply simple opening/closing operators with a small structuring element to suppress small local perturbations while preserving the main sign structure. For FFT-based denoising, we will transform the image to the frequency domain, attenuate suspicious high-frequency components, and reconstruct the image.

4 Deliverables

4.1 Must accomplish

1. Train a baseline traffic sign classifier on GTSRB and report clean accuracy.
2. Implement white-box FGSM and PGD attacks and evaluate adversarial accuracy and attack success rate.
3. Compare attack success rate using at least two defense methods, including preprocessing and adversarial training.

4.2 Expect to accomplish

1. Implement and evaluate the C&W attack.
2. Design and evaluate the custom edge-aware attack.
3. Implement a gray-box setting using a surrogate or parameter-mismatched model and compare it with the white-box setting.

4.3 Would like to accomplish

1. Evaluate morphology-based denoising and FFT-based denoising as custom defenses.
2. Apply AutoAttack for a final robustness benchmark.
3. Highlight how different defenses perform better against specific adversarial attacks or vice versa.

References

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